

# 1D CNN — Two-Hand Dynamic Gesture Run Report

Generated 2026-05-07 22:51:44

## Data paths

|                      |                                                                   |
|----------------------|-------------------------------------------------------------------|
| Training data path 1 | /home/jestin/ThesisRepo/ML/NewTestData/Alan/Dynamic · 30 trials   |
| Training data path 2 | /home/jestin/ThesisRepo/ML/NewTestData/Harry/Dynamic · 30 trials  |
| Training data path 3 | /home/jestin/ThesisRepo/ML/NewTestData/Bella/Dynamic · 30 trials  |
| Training data path 4 | /home/jestin/ThesisRepo/ML/NewTestData/Jestin/Dynamic · 24 trials |
| Training data path 5 | /home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Dynamic · 5 trials |
| Training data path 6 | /home/jestin/ThesisRepo/ML/NewTestData/Marcus/Dynamic · 30 trials |
| Training data path 7 | /home/jestin/ThesisRepo/ML/NewTestData/Tash/Dynamic · 30 trials   |
| Total training paths | 7 · 179 trials total                                              |
| Batch-test data root | /home/jestin/ThesisRepo/ML/NewTestData/Anghad/Dynamic             |
| Saved model path     | saved_models/cnn_model_2026-05-07_22-51-44_5fadc33dd126.keras     |
| Experiment log       | saved_models/experiment_results.csv                               |

## Sensor selection

|          |                                                |
|----------|------------------------------------------------|
| Hands    | left, right                                    |
| Segments | wrist, palm, thumb, index, middle, ring, pinky |
| Features | YPR, Accel, Flex                               |
| Channels | 176 (sequence length 90)                       |

## Preprocessing & split

|                  |                                                                                             |
|------------------|---------------------------------------------------------------------------------------------|
| Resample steps   | 90                                                                                          |
| Butterworth      | on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz                                                  |
| Normalisation    | minmax                                                                                      |
| Test size / seed | 0.2 · random_state=42 · stratify=True                                                       |
| Sample counts    | total=179 · train pool=143 · train+aug=429 · hold-out=36                                    |
| Classes          | 6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll |

## Augmentation (training only)

|                      |                                          |
|----------------------|------------------------------------------|
| Enabled              | yes                                      |
| Copies per sample    | 2                                        |
| Random seed          | 42                                       |
| Active augmentations | amplitude_scale, baseline_offset         |
| • amplitude_scale    | apply_prob=0.5, scale_range=(0.95, 1.05) |
| • baseline_offset    | apply_prob=0.5, offset_std_ratio=0.02    |

## Model architecture & training

|                                 |                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Architecture                    | Conv1D(32, k=4, ReLU) → MaxPool(2) → Conv1D(64, k=4, ReLU) → MaxPool(2) → Flatten → Dense(64, ReLU) → Dropout(0.3) → Dense(softmax) |
| Optimizer / loss                | adam · sparse_categorical_crossentropy                                                                                              |
| Input shape                     | (90, 176)                                                                                                                           |
| Trainable params                | 113,190                                                                                                                             |
| Epochs · batch size · val split | 15 · 16 · 0.2                                                                                                                       |

## Cross-validation

|                    |                 |
|--------------------|-----------------|
| Folds · shuffle    | 5 · True        |
| Mean ± std val acc | 0.9860 ± 0.0172 |
| Best fold          | 1               |

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 342              | 29  | 1.0000  | 1.0000   | 1.0000    |
| 2    | 342              | 29  | 1.0000  | 1.0000   | 1.0000    |
| 3    | 342              | 29  | 0.9655  | 0.9655   | 0.9655    |
| 4    | 345              | 28  | 0.9643  | 0.9643   | 0.9643    |
| 5    | 345              | 28  | 1.0000  | 1.0000   | 1.0000    |

## Final training & hold-out test

|                                |                 |
|--------------------------------|-----------------|
| Final train acc / loss         | 0.9796 · 0.0685 |
| Final val acc / loss           | 0.9884 · 0.0568 |
| Best val acc / lowest val loss | 1.0000 · 0.0214 |
| Hold-out test acc / loss       | 0.9722 · 0.0958 |

## Per-class hold-out metrics

| Class        | Precision | Recall | F1    | Support |
|--------------|-----------|--------|-------|---------|
| Double_Lower | 1.000     | 1.000  | 1.000 | 6       |

|                      |       |       |       |    |
|----------------------|-------|-------|-------|----|
| Double_Pistol_Recoil | 1.000 | 1.000 | 1.000 | 6  |
| Double_Raise         | 1.000 | 1.000 | 1.000 | 6  |
| Double_Wiggle        | 1.000 | 0.833 | 0.909 | 6  |
| Double_cmere         | 1.000 | 1.000 | 1.000 | 7  |
| Drum_Roll            | 0.833 | 1.000 | 0.909 | 5  |
| macro avg            | 0.972 | 0.972 | 0.970 | 36 |
| weighted avg         | 0.977 | 0.972 | 0.972 | 36 |

Batch test (NewTestData)

| Class                | Correct | Total | Accuracy |
|----------------------|---------|-------|----------|
| Double_Lower         | 5       | 5     | 1.000    |
| Double_Pistol_Recoil | 5       | 5     | 1.000    |
| Double_Raise         | 5       | 5     | 1.000    |
| Double_Wiggle        | 2       | 5     | 0.400    |
| Double_cmere         | 6       | 6     | 1.000    |
| Drum_Roll            | 5       | 5     | 1.000    |
| Overall              | 28      | 31    | 0.903    |